

Myungjoon Kim

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Education

- **Korea Advanced Institute of Science and Technology (KAIST)** **Daejeon, Korea**
Ph.D. , Materials Science and Engineering *Sep 2018 - Feb 2023*
Thesis : Inverse design of free-form metasurfaces: from adjoint-based optimization to physics-informed deep learning (*Advisor : Prof. Jonghwa Shin*)
- **Korea Advanced Institute of Science and Technology (KAIST)** **Daejeon, Korea**
M.S., Materials Science and Engineering *Mar 2015 - Feb 2017*
- **Korea Advanced Institute of Science and Technology (KAIST)** **Daejeon, Korea**
B.S., Materials Science and Engineering, Computer Science (Double major) *Feb 2008- Feb 2015*
*incl. military service as a Korean Augmentation To the United States Army (KATUSA), 2010–2012

Research Interests

Nonlinear multimode photonics, Inverse design, AI for Physics.

Research Experience

- **Cornell University** **Dec 2024 –**
Schmidt AI in Science Postdoc (PI: Prof. Frank Wise) **Ithaca, USA**
- **Korea Advanced Institute of Science and Technology (KAIST)** **Apr 2023 – Nov 2024**
Postdoctoral Researcher (PI: Prof. Jonghwa Shin) **Daejeon, Korea**
- **Computational Science Research Center, KIST** **Jan 2018 – Aug 2018**
Research Intern (Mentor: Dr. Donghun Kim) **Seoul, Korea**
 - Develop deep learning models for predicting adsorption energy of catalysis
- **Interactive Computing Lab, KAIST** **Oct 2014 – Feb 2015**
Research Intern (Advisor : Prof. Uichin Lee) **Daejeon, Korea**
 - Participated in projects on detecting capture moment and correcting device orientation of mobile phone

Selected Projects

- **Nonlinear multimode photonics**
 - Implemented differentiable GMMNLSE and (3+1)D GNLSE solvers in PyTorch, then developed physics-informed neural operators in the guided eigenmode basis of multimode fibers for spatiotemporal pulse propagation.

- Applied the framework to inverse design of input fields, targeting spatial, temporal, and spectral properties at the output.
- **Inverse design**
 - Developed adjoint-based optimization frameworks for free-form metasurfaces, designing phase masks for 3D photolithography.
 - Proposed a field optimization algorithm based on ADMM, achieving higher-performing structures than topology optimization across a range of photonic design problems.
- **AI for physics**
 - Built equivariant neural surrogate solvers for Maxwell's equations, with translation and flip equivariance of metasurfaces guaranteed by architecture rather than data augmentation.
 - Explored large language models for nanophotonic inverse design, testing the feasibility of automating design workflows without domain expertise.

Publications

('†' indicates equal contribution)

- V Sharma, N Bender, H Haig, **M Kim**, DN Christodoulides, FW Wise*, "Observation of stability of Gaussian beams and off-axis beam-cleaning in graded-index rods", [Optics Express](#), 33, 48130 (2025)
- JR Lee†, **M Kim**†, SH Moon, DS Choi, J Shin, IS Kang, TJ Je*, E Jeon*, "Manufacturing of a bitmap-type lightweight microwave absorber metasurface", [International Journal of Precision Engineering and Manufacturing](#), 26, 3461 (2025)
- **M Kim**†, H Park†, J Shin*, "Nanophotonic device design based on large language models: multilayer and metasurface examples", [Nanophotonics](#), 14, 1273 (2025)
- D Lee, M Lee, B Yerenzhap, **M Kim**, H Hugonnet, S Jeon, J Shin, Y Park, "Fast free-form phase mask design for three-dimensional photolithography using convergent Born series", [ACS Photonics](#), 12, 610 (2024)
- M Cho, J Jung, **M Kim**, J Lee, S Min, J Hong, S Lee, M Heo, JU Kim, I-S Joe, J Shin*, "Color arrestor pixels for high-fidelity, high-sensitivity imaging sensors", [Nanophotonics](#), 13, 2971 (2024)
- **M Kim**†, N Kim†, J Shin*, "Realization of all two-dimensional Bravais lattices with metasurface-based interference lithography", [Nanophotonics](#), 13, 1467 (2024)
- N Kim, **M Kim**, J Jung, T Chang, S Jeon, J Shin*, "Highly angle-sensitive and efficient optical metasurfaces with broken mirror symmetry", [Nanophotonics](#), 12, 2347 (2023)
- A Baucour, **M Kim**, J Shin*, "Data-driven concurrent nanostructure optimization based on conditional generative adversarial networks", [Nanophotonics](#), 11, 2865 (2022)
- SH Nam†, **M Kim**†, N Kim†, D Cho, M Choi, JH Park, J Shin*, S Jeon*, "Photolithographic realization of target nanostructures in 3D space by inverse design of phase modulation", [Science Advances](#), 8, abm6310 (2022)
- S Hong*, CH Liow, JM Yuk, HR Byon, Y Yang, EA Cho, J Yeom, G Park, H Kang, S Kim, Y Shim, M Na, C Jeong, G Hwang, H Kim, H Kim, S Eom, S Cho, H Jun, Y Lee, A Baucour, K Bang, **M Kim**, S Yun, J Ryu, Y Han, A Jettybayeva, P-P Choi, J C Agar, S V Kalinin, P W Voorhees, P Littlewood, H M Lee, "Reducing

time to discovery: materials and molecular modeling, imaging, informatics, and integration", [ACS Nano](#), **15**, 3971 (2021)

- N Kim, T Chang, **M Kim**, M Heo, A Baucour, J Jung, J Shin*, "Spectrally sharp metasurfaces for wide-angle high extinction of green lasers", [Optics Express](#), **28**, 22121 (2020)
- **M Kim**†, BC Yeot†, Y Park, HM Lee, SS Han*, D Kim*, "Artificial Intelligence to Accelerate the Discovery of N2 Electroreduction Catalysts", [Chemistry of Materials](#), **32**, 709 (2020)
- H Kim, **M Kim**, T Chang, A Baucour, S Jeon, N Kim, H-J Choi, H Lee, and J Shin*, "Bright and vivid plasmonic color filters having dual resonance modes with proper orthogonality", [Optics Express](#), **26**, 27403 (2018)
- J Oh, J Kim, **M Kim**, W Choi, S Lee, U Lee*, "Understanding Mobile Document Capture and Correcting Orientation Errors", [International Journal of Human-Computer Studies](#), **104**, 64 (2017)

Conferences and Presentations

- **M Kim**, V Sharma, N. Bender, F. Wise, "Inverse design for spatiotemporal control of ultrafast light in optical fiber via physics-informed machine learning", SPIE Optics + Photonics (2026)
- (Invited Talk) **M Kim**, "Optimizing Photonic and Material Design with Differentiable Programming", Guest Lecture, AI for Materials, Cornell University (2025)
- (Invited Talk) **M Kim**, J Park, J Shin, "Toward accurate neural Maxwell solvers for inverse design of metasurfaces", KPS Spring meeting (2024)
- **M Kim**, J. Park, J Shin, "Efficient Fourier neural operators by group convolution and channel shuffling", ICLR Workshop on AI4DifferentialEquations In Science (2024)
- **M Kim**, N Kim, J Shin, "Concurrent Inverse Design of Structured Light and Metasurface for Nanopatterning Process", Frontier in Optics (2022)

Skills

- **Simulations:** FDTD (Tidy3D, Lumerical, Meep), RCWA, DDA, GMMNLSE / (3+1)D GNLSE
- **Machine learning:** PyTorch, Jax

Academic Service

- **Reviewer:** ICLR, NeurIPS

Reference

- **Jonghwa Shin**, Professor (qubit@kaist.ac.kr)
Department of Materials Science and Engineering, KAIST
- **Frank Wise**, Professor (frank.wise@cornell.edu)
School of Applied and Engineering Physics, Cornell University
- **Owen D. Miller**, Associate Professor (owen.miller@yale.edu)
Department of Applied Physics and Energy Sciences Institute, Yale University

- **Donghun Kim**, Associate Professor (donghun.kim@kaist.ac.kr)
Department of Materials Science and Engineering, KAIST